WAP to Implement Singly Linked List with following operations
a) Create a linked list.
b) Deletion of first element, specified element and last element in the list.
c) Display the contents of the linked list.

```
#include <stdio.h>
#include <stdlib.h>
struct node {
    int data;
    struct node *next;
};
struct node *head = NULL;
void createList(int n) {
    struct node *newNode, *temp;
    int data, i;
    if (n <= 0) {
        printf("Number of nodes should be greater than 0.\n");
        return;
    }
    for (i = 1; i <= n; i++) {
        newNode = (struct node*)malloc(sizeof(struct node));
        if (newNode == NULL) {
            printf("Memory allocation failed.\n");
            return;
        }
        printf("Enter data for node %d: ", i);
        scanf("%d", &data);
        newNode->data = data;
        newNode->next = NULL;
        if (head == NULL) {
            head = newNode;
        } else {
            temp->next = newNode;
        }
        temp = newNode;
    }
    printf("\nLinked list created successfully.\n");
}
void deletefirst(){
  struct node *temp;
  if(head==NULL){
       printf("list is empty.nothing to delete.\n");
```

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        return;
        }
        temp=head;
        head=head->next;
        printf("deleted element:%d\n",temp->data);
        free(temp);
}
void deletelast() {
    struct node *temp, *prev;
    if (head == NULL) {
        printf("List is empty. Nothing to delete.\n");
        return;
    }
    if (head->next == NULL) {
        printf("Deleted element: %d\n", head->data);
        free(head);
        head = NULL;
        return;
    }
    temp = head;
    while (temp->next != NULL) {
        prev = temp;
        temp = temp->next;
    }
    printf("Deleted element: %d\n", temp->data);
    prev->next = NULL;
    free(temp);
}
void deletespecific(int value) {
    struct node *temp = head,*prev = NULL;
    if (head == NULL) {
        printf("List is empty. Nothing to delete.\n");
        return;
    }
    if (head->data == value) {
        temp = head;
        head = head->next;
        printf("Deleted element: %d\n", temp->data);
        free(temp);
        return;
    }
    while (temp != NULL && temp->data != value) {
        prev = temp;
        temp = temp->next;
```

```
    }
    if (temp == NULL) {
        printf("Element %d not found in the list.\n", value);
        return;
    }
    prev->next = temp->next;
    printf("Deleted element: %d\n", temp->data);
    free(temp);
}
void displaylist() {
    struct node *temp = head;
    if (head == NULL) {
        printf("List is empty.\n");
        return;
    }
    printf("\nLinked List: ");
    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
int main() {
    int choice, n,value;

    while (1) {
        printf("\n--- Singly Linked List Operations ---\n1. Create Linked List\n2. delete first\n3.delete
specific\n4.delete end\n5. Display List\n6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
    case 1:
        printf("Enter number of nodes: ");
        scanf("%d", &n);
        createList(n);
        break;
    case 2:
        deletefirst();
        break;
    case 3:
        printf("Enter value to delete: ");
        scanf("%d", &value);
        deletespecific(value);
        break;
```

```
    case 4:
        deletelast();
        break;
    case 5:
        displaylist();
        break;
    case 6:
        printf("Exiting...\n");
        exit(0);
    default:
        printf("Invalid choice. Try again.\n");
    }
}
return 0;
}
```

```
--- Singly Linked List Operations ---
1. Create Linked List
2. delete first
3.delete specific
4.delete end
5. Display List
6. Exit
Enter your choice: 1
Enter number of nodes: 5
Enter data for node 1: 10
Enter data for node 2: 20
Enter data for node 3: 30
Enter data for node 4: 40
Enter data for node 5: 50

Linked list created successfully.

--- Singly Linked List Operations ---
1. Create Linked List
2. delete first
3.delete specific
4.delete end
5. Display List
6. Exit
Enter your choice: 2
deleted element:10

--- Singly Linked List Operations ---
1. Create Linked List
2. delete first
3.delete specific
4.delete end
5. Display List
6. Exit
Enter your choice: 3
Enter value to delete: 40
Deleted element: 40

--- Singly Linked List Operations ---
1. Create Linked List
2. delete first
3.delete specific
4.delete end
5. Display List
6. Exit
Enter your choice: 4
Deleted element: 50
```

```
--- Singly Linked List Operations ---
1. Create Linked List
2. delete first
3.delete specific
4.delete end
5. Display List
6. Exit
Enter your choice: 5

Linked List: 20 -> 30 -> NULL

--- Singly Linked List Operations ---
1. Create Linked List
2. delete first
3.delete specific
4.delete end
5. Display List
6. Exit
Enter your choice: 6
Exiting...

Process returned 0 (0x0)   execution time : 46.144 s
Press any key to continue.
```